

ClaimGuard Project Requirements and Wireframes

BUAN 3301 AI in Business Meridian Insurance course prototype

ClaimGuard is a working, single-page claim-review application. It associates each instructor-provided claim with two photos, checks photo reuse within the claim set and on the public web, applies a deterministic decision rule, and prepares a Google Form for human submission. The system stops before email collection and Submit; a signed-in person completes that step.

Purpose and scope

The primary user is a team member or adjuster reviewing a claim. The page supports two entry modes: selecting one of the 18 seeded claims or entering a new claim with photos. The design is an end-user workflow, not an administration dashboard. Meridian Insurance is fictional, and the result is a course prototype rather than a finding of fraud.

End to end user flow

1. Select a mapped reference claim or enter a new claim description and factual fields.
2. Review two mapped thumbnails, or add two to six new photos by picker, drag and drop, or ZIP.
3. Run one web lookup per image. The app compares each photo with other claims and reports web matches or an unavailable check.
4. The server rebuilds the evidence, applies the guardrail, and displays a timeline, photo evidence, and the decision.
5. Open a pre-filled Google Form containing claim ID, claimant, lookup summary, and decision. A real person signs in, lets the Form collect their email, and clicks Submit.

Decision boundary

A photo match is a reason for human review, not proof of deception. An unavailable web check escalates rather than silently passing. The Google Form email is never fabricated or pre-filled, and the application has no form-submission action.

Prepared with AI assistance and checked against the implemented ClaimGuard application and the instructor project brief.

Functional requirements

The following requirements describe the current application. IDs provide a concise checklist for the team demo.

ID	Requirement and observable behavior
R1	The team manually maps every one of 18 claims to two distinct reference JPGs. Validation rejects missing, duplicate, or unknown filenames and requires all 36 photos exactly once. No model guesses mappings.
R2	The searchable existing-claim picker filters by claim ID or claimant. Selection shows incident facts, narrative, customer email, and the two mapped thumbnails; unmapped claims cannot run.
R3	New claims require claimant, incident date, positive USD amount no greater than \$250,000, and description. Claim ID is optional and can be generated; location, category, and customer email are optional. TXT, MD, and DOCX description imports are supported.
R4	New claims accept two to six JPEG, PNG, or WebP photos. Client-side ZIP intake ignores junk and non-images, allows at most 50 entries and 30 MB expanded, rejects nested archives, and lets users remove extras. Each lookup request carries one image, capped at 4 MB.
R5	Google Cloud Vision Web Detection receives base64 image bytes. Results distinguish full, partial, and visually similar matches, show source and time, and use a SHA-256 cache. A failed check is labeled unavailable, never zero matches.
R6	The server compares photos across different claims using exact SHA-256 or dHash visual similarity. The current near-match cutoff is Hamming distance 6; matches show the other claim and photo.
R7	The server, not the browser, rebuilds evidence and applies one deterministic guardrail. Missing photos block a decision; reuse, qualifying non-stock web matches, or unavailable web checks escalate; contextual narrative similarities remain notes.
R8	Results show an agent-activity timeline, one evidence card per photo, a decision banner, reasons, notes, source badges, cross-claim matches, an optional adjuster notice, and a live recheck action.
R9	The Google Form has Collect email addresses enabled. The link pre-fills exactly claim ID, claimant name, a photo-only lookup summary of at most 300 characters, and Auto-approve or Escalate. It never includes an email entry or submits the form.
R10	MongoDB holds claims, photo evidence, cached lookups, decisions, audit history, and daily usage counts. User-origin data expires after 14 days; uploaded originals are not stored. Optional LLM features never control the decision.

Decision rules and safeguards

Current thresholds are versioned placeholders for team review. A reference photo set validation is available, but the team should justify any final threshold using the real images and the false-approval versus false-escalation tradeoff.

Situation	Current handling
Fewer than two photos	Blocked; no completed decision or form link.
Both photos reused across claims	Escalate; exact or near-duplicate matches are explained.
One photo reused across claims	Escalate under the current partial-match setting.
At least three non-stock full web matches	Escalate; stock-source image hosts do not count toward three.
Web check unavailable with no usable result	Escalate for human review.
Similar narrative, same claimant, or own-photo similarity	Record a note; these signals alone do not escalate.

The stock-source allowlist currently contains pexels.com, pixabay.com, and unsplash.com, including their subdomains. A site outside that list is not blocked or accused; a full-image match there simply counts toward the threshold. Partial and visually similar web results are shown for context but do not count as full matches.

Operational and privacy requirements

- Each API request performs a small unit of work within the Node route time budget. The browser unpacks ZIP files and never posts a complete ZIP or batch of photos to the server.
- Secrets remain in server-side environment variables. Reference originals are outside the public web directory; only stored thumbnails have an image endpoint. The page requests noindex and disallows crawlers.
- A retryable database failure is retried briefly, then shown with a manual Try again action. Vision errors are distinct from genuine clean results. Daily API limits and a best-effort lookup rate limit reduce uncontrolled service use.
- The interface supports keyboard use, readable status messages, responsive layout, result printing, and a Start Over action that clears only on-screen state.

Demonstration acceptance checks

A successful course demonstration should show: the validated 18-to-36 mapping; one real photo lookup, including a reused-image finding; one claim run from selection through the decision timeline; an unavailable-check escalation; and the pre-filled Form with the submitter email left to a real person. A zero-LLM configuration must still work. The team should show the actual Form setting and its own final threshold justification separately.

Entry screen wireframes

The following low-fidelity layouts show the desktop reading order. On a narrow screen, side-by-side cards stack vertically while the same controls remain available.

MERIDIAN INSURANCE **ClaimGuard**

Select existing claim **Submit new claim**

Search claim ID or claimant name

Claim details
Date | location | category | amount | customer email
Claim narrative

Photo 1
thumbnail

Photo 2
thumbnail

Run ClaimGuard check

Figure 1 Existing claim mode. Search and claim details precede the two mapped photos and bottom action.

MERIDIAN INSURANCE **ClaimGuard**

Select existing claim **Submit new claim**

Claim ID optional

Claimant name required

Incident date required

Claim amount required

Optional location category customer email

Description plus TXT MD or DOCX import

Drop photos here or choose files and ZIP
Queue shows preview | status | remove 2 to 6 photos

Run ClaimGuard check

Figure 2 New claim mode. Factual inputs and description come before the photo queue and bottom action.

Result screen wireframe

After lookups finish, the result area appears below the entry form. It shows what the agent did and why the guardrail reached its outcome before presenting the human-only Form handoff.

MERIDIAN INSURANCE **ClaimGuard**

Your claim review

Agent activity

- Claim loaded
- Photo 1 checked
- Photo 2 checked
- Rule applied
- Form link prepared
- Human submission pending

Photo 1 evidence

thumbnail | Live or Cached

Web counts and domains

Cross claim matches

Re run live lookup

Photo 2 evidence

thumbnail | Live or Cached

Web counts and domains

Cross claim matches

Re run live lookup

Decision banner | Reasons and notes

Open pre filled form | Copy summary | Save as PDF | Start Over

Human signs in and clicks Submit | Agent never submits

Figure 3 Result area. Evidence and the agent timeline lead to the decision, explanation, and human actions.

Interaction notes

- An evidence card separates web matches, a verified clean web result, and an unavailable check with different messages. Re-running live invalidates the old decision until the new evidence is evaluated.
- The Form button opens a new tab for the human. Copy summary, Save as PDF, and Start Over act on the screen; Start Over does not delete database records.
- The activity timeline always leaves the final human sign-in and Submit step pending. Optional AI wording may help an adjuster read the result but cannot alter it.